VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT
on

Object Oriented Java Programming

(23CS3PCOOJ)

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
BENGALURU-560019
Aug-2025 to Jan-2026

B.M.S. College of Engineering,

Bull Temple Road, Bangalore 560019

(Affiliated To Visvesvaraya Technological University, Belgaum)

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Object Oriented Java Programming (23CS3PCOOJ)" carried out by **Riya Ann Mathews (1BF24CS254),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of an Object-Oriented Java Programming (23CS3PCOOJ) work prescribed for the said degree.

Dr. Seema Patil Associate Professor Department of CSE, BMSCE	Dr. Kavitha Sooda Professor & HOD Department of CSE, BMSCE

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Program 1:

Implement Quadratic Equation

C: > Users > Riya > Desktop > 1BF24CS254 > Quadratic.java > Quadratic

```java
import java.util.Scanner;
class Quadratic
{
    public static void main (String args[])
    {
        int a,b,c,d;
        double r1,r2;
        Scanner in=new Scanner(System.in);
        System.out.println("Enter coefficients of quadratic equation");
        a=in.nextInt();
        b=in.nextInt();
        c=in.nextInt();
        if(a==0||b==0||c==0)
        {
            System.out.println("Not a quadratic equation");
        }
        else
        {
            d=(b*b)-(4*a*c);
            if(d==0)
            {
                System.out.println("Roots are real and equal");
                r1=(-b/(2*a));
                System.out.println("Root="+r1);
            }
            else if(d>0)
            {
                System.out.println("Roots are real and distinct");
                r1=((-b)+(Math.sqrt(d)))/(double)(2*a);
                r2=((-b)-(Math.sqrt(d)))/(double)(2*a);
                System.out.println("Root1="+r1);
                System.out.println("Root2="+r2);


            }
            else if(d<0)
```

C: > Users > Riya > Desktop > 1BF24CS254 > Quadratic.java > Quadratic

```java
class Quadratic
    public static void main (String args[])
            d=(b*b)-(4*a*c);
            if(d==0)
            {
                System.out.println(x: "Roots are real and equal");
                r1=(-b/(2*a));
                System.out.println("Root="+r1);
            }
            else if(d>0)
            {
                System.out.println(x: "Roots are real and distinct");
                r1=((-b)+(Math.sqrt(d)))/(double)(2*a);
                r2=((-b)-(Math.sqrt(d)))/(double)(2*a);
                System.out.println("Root1="+r1);
                System.out.println("Root2="+r2);


            }
            else if(d<0)
            {
                System.out.println(x: "Roots are imaginary");
                r1=(-b/(double)(2*a));
                r2=(Math.sqrt(Math.abs(d)))/(2*a);
                System.out.println("Root1="+r1+"+i"+r2);
                System.out.println("Root2="+r1+"-i"+r2);

            }
            else
            {
                System.out.println(x: "Invalid input");
            }
        }
    }
}
```

Output:

```
PS C:\Users\Admin\Desktop\1BF24CS254>  & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Admin\AppData\Roaming\Code\User\workspaceStorage\099b02178a966e8eb1514e4018670ad1\redhat.java\jdt_ws\1BF24CS254_37831adc\bin' 'Quadratic'
Enter coefficients of quadratic equation
0
9
8
Not a quadratic equation
PS C:\Users\Admin\Desktop\1BF24CS254> ^C
PS C:\Users\Admin\Desktop\1BF24CS254>  c:; cd 'c:\Users\Admin\Desktop\1BF24CS254'; & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExcepti
Enter coefficients of quadratic equation
1
4
2
Roots are real and distinct
Root1=-0.5857864376269049
Root2=-3.414213562373095
PS C:\Users\Admin\Desktop\1BF24CS254>
ratic'
Enter coefficients of quadratic equation
2
4
2
Roots are real and equal
Root=-1.0
PS C:\Users\Admin\Desktop\1BF24CS254> ^C
PS C:\Users\Admin\Desktop\1BF24CS254>  c:; cd 'c:\Users\Admin\Desktop\1BF24CS254'; & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExcepti
Enter coefficients of quadratic equation
1
2
3
Roots are imaginary
Root1=-1.0+i1.4142135623730951
Root2=-1.0-i1.4142135623730951
PS C:\Users\Admin\Desktop\1BF24CS254> ^C
PS C:\Users\Admin\Desktop\1BF24CS254>
```

Program 2:

Implement SGPA calculator

C: > Users > Riya > Desktop > 1BF24CS254 > Main.java > ...

```java
import java.util.Scanner;

class Subject {
    int subjectMarks;
    int grade;
    int credits;
}

class Student {
    String name, usn;
    double SGPA;
    Subject subjects[];
    Scanner s;
    Student() {
        s = new Scanner(System.in);
        subjects = new Subject[8];
        for (int i = 0; i < 8; i++) {
            subjects[i] = new Subject();
        }
    }
    void getStudentDetails() {
        System.out.print(s: "Enter Student Name: ");
        name = s.nextLine();
        System.out.print(s: "Enter Student USN: ");
        usn = s.nextLine();
    }
    void getMarks() {
        for (int i = 0; i < 8; i++) {
            System.out.println("\nEnter details for Subject " + (i + 1));
            System.out.print(s: "Marks (out of 100): ");
            subjects[i].subjectMarks = s.nextInt();
            System.out.print(s: "Credits: ");
            subjects[i].credits = s.nextInt();
            if (subjects[i].subjectMarks > 100 || subjects[i].subjectMarks < 0) {
                System.out.println(x: "Invalid marks entered! Setting to 0.");
                subjects[i].subjectMarks = 0;
```

0 ⚠ 1 Java: Ready Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} Java Signed out Go Live

C: > Users > Riya > Desktop > 1BF24CS254 > Main.java > ...

```java
class Student {
    void getMarks() {
                subjects[i].subjectMarks = 0;
            }
            subjects[i].grade = subjects[i].subjectMarks / 10;
            if (subjects[i].grade == 11) {
                subjects[i].grade = 10;
            } else if (subjects[i].grade <= 4) {
                subjects[i].grade = 0;
            }
        }
    }
    void computeSGPA() {
        int effectiveScore = 0, totalCredits = 0;
        for (int i = 0; i < 8; i++) {
            effectiveScore += subjects[i].grade * subjects[i].credits;
            totalCredits += subjects[i].credits;
        }
        if (totalCredits > 0)
            SGPA = (double) effectiveScore / totalCredits;
        else
            SGPA = 0.0;
    }
    void displayResult() {
        System.out.println(x: "\n===== STUDENT RESULT =====");
        System.out.println("Name: " + name);
        System.out.println("USN: " + usn);
        System.out.printf(format: "SGPA: %.2f\n", SGPA);
    }
}
public class Main {
    Run | Debug
    public static void main(String args[]) {
        Student st = new Student();
        st.getStudentDetails();
        st.getMarks();
```

0 ⚠ 1 Java: Ready Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} Java Signed out Go Live

C: > Users > Riya > Desktop > 1BF24CS254 > Main.java > ...

```java
class Student {
    void getMarks() {
                subjects[i].grade = 10;
            } else if (subjects[i].grade <= 4) {
                subjects[i].grade = 0;
            }
        }
    }
    void computeSGPA() {
        int effectiveScore = 0, totalCredits = 0;
        for (int i = 0; i < 8; i++) {
            effectiveScore += subjects[i].grade * subjects[i].credits;
            totalCredits += subjects[i].credits;
        }
        if (totalCredits > 0)
            SGPA = (double) effectiveScore / totalCredits;
        else
            SGPA = 0.0;
    }
    void displayResult() {
        System.out.println("\n===== STUDENT RESULT =====");
        System.out.println("Name: " + name);
        System.out.println("USN: " + usn);
        System.out.printf("SGPA: %.2f\n", SGPA);
    }
}
public class Main {
    public static void main(String args[]) {
        Student st = new Student();
        st.getStudentDetails();
        st.getMarks();
        st.computeSGPA();
        st.displayResult();
    }
}
```

Output:

```
PS C:\Users\Admin\Desktop\1BF24CS254>  & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Admin\AppData\Roaming\Code\User\workspaceStorage\099b02178a966e8eb1514e4018670ad1\redhat.java\jdt_ws\1BF24CS254_37831adc\bin' 'Main'
Enter number of students:
2
Enter Student Name: Diya
Enter Student USN: 1BF24CS204
Enter marks for Subject1:
95
Enter credits for Subject1:4
Enter marks for Subject2:
92
Enter credits for Subject2:4
Enter marks for Subject3:
88
Enter credits for Subject3:3
Enter marks for Subject4:
80
Enter credits for Subject4:3
Enter marks for Subject5:
85
Enter credits for Subject5:3
Enter marks for Subject6:
70
Enter credits for Subject6:1
Enter marks for Subject7:
67
Enter credits for Subject7:1
Enter marks for Subject8:
83
Enter credits for Subject8:1

Student result:
Name: Diya
USN: 1BF24CS204
SGPA: 9.25
Enter Student Name: Aparna
Enter Student USN: 1BF24CS254
Enter marks for Subject1:
100
```

```
Enter marks for Subject1:
100
Enter credits for Subject1:4
Enter marks for Subject2:
97
Enter credits for Subject2:4
Enter marks for Subject3:
89
Enter credits for Subject3:3
Enter marks for Subject4:
82
Enter credits for Subject4:3
Enter marks for Subject5:
90
Enter credits for Subject5:3
Enter marks for Subject6:
70
Enter credits for Subject6:1
Enter marks for Subject7:
86
Enter credits for Subject7:1
Enter marks for Subject8:
100
Enter credits for Subject8:1

Student result:
Name: Aparna
USN: 1BF24CS254
SGPA: 9.55
PS C:\Users\Admin\Desktop\1BF24CS254>
```

Program 3:

Implement toString() function

C: > Users > Riya > Desktop > 1BF24CS254 > Bookdetails.java

```java
import java.util.Scanner;
class Books
{
    String name;
    String author;
    int price;
    int numPages;
    Books(String name,String author,int price,int numPages)
    {
        this.name=name;
        this.author=author;
        this.price=price;
        this.numPages=numPages;

    }
    public String toString()
    {
        String name,author,price,numPages;
        name="Book name:"+this.name+"\n";
        author="Author name:"+this.author+"\n";
        price="Price:"+this.price+"\n";
        numPages="Number of pages:"+this.numPages+"\n";
        return name+author+price+numPages;

    }
}
public class Bookdetails
{
    public static void main(String args[])
    {
        Scanner s=new Scanner(System.in);
        int n,price,numPages;
        String name,author;
        System.out.println("Enter no of books:");
        n=s.nextInt();
        s.nextLine();
```

0 ⚠ 0 Java: Activating... Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Java Signed out Go Live

C: > Users > Riya > Desktop > 1BF24CS254 > Bookdetails.java > ...

```java
public class Bookdetails
{
    Run | Debug
    public static void main(String args[])
    {
        Scanner s=new Scanner(System.in);
        int n,price,numPages;
        String name,author;
        System.out.println(x: "Enter no of books:");
        n=s.nextInt();
        s.nextLine();
        Books b[];
        b=new Books[n];
        for(int i=0;i<n;i++)
        {
            System.out.println(x: "Enter name of book:");
            name=s.nextLine();
            System.out.println(x: "Enter author of book:");
            author=s.nextLine();
            System.out.println(x: "Enter price of book:");
            price=s.nextInt();
            s.nextLine();
            System.out.println(x: "Enter no of pages of book:");
            numPages=s.nextInt();
            s.nextLine();
            b[i]=new Books(name,author,price,numPages);
            System.out.println("Details of book " + (i + 1) + ":\n" + b[i].toString());


        }


    }
}
```

Output:

```
PS C:\Users\Admin\Desktop\1BF24CS254> cd "c:\Users\Admin\Desktop\1BF24CS254\" ; if ($?) { javac Bookdetails.java } ; if ($?) { java Bookdetails }
Enter no of books:
2
Enter name of book:
Pride and Prejudice
Enter author of book:
Jane Austen
Enter price of book:
350
Enter no of pages of book:
500
Details of book 1:
Book name:Pride and Prejudice
Author name:Jane Austen
Price:350
Number of pages:500

Enter name of book:
The Great Gatsby
Enter author of book:
F Scott Fitzergald
Enter price of book:
300
Enter no of pages of book:
650
Details of book 2:
Book name:The Great Gatsby
Author name:F Scott Fitzergald
Price:300
Number of pages:650

PS C:\Users\Admin\Desktop\1BF24CS254>
```

Program 4:

Implement Abstract class

C: > Users > Riya > Downloads > AbstractArea.java > ...

```java
import java.util.Scanner;
abstract class Shape
{
    double dim1,dim2;
    Scanner s=new Scanner(System.in);
    abstract void input();
    abstract void printArea();
}
class Rectangle extends Shape
{
    void input()
    {
        System.out.println(x: "Enter dimensions of Rectangle(length and breadth):");
        dim1=s.nextDouble();
        dim2=s.nextDouble();

    }
    void printArea()
    {
        System.out.println("Area of Rectangle:"+(dim1*dim2));
    }
}
class Triangle extends Shape
{
    void input()
    {
        System.out.println(x: "Enter dimensions of Triangle(base and height):");
        dim1=s.nextDouble();
        dim2=s.nextDouble();
    }
    void printArea()
    {
        System.out.println("Area of Triangle:"+(0.5*dim1*dim2));
    }
}
class Circle extends Shape
```

0 ⚠ 1 Java: Ready Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Java Signed out Go Live

C: > Users > Riya > Downloads > AbstractArea.java > ...

```java
class Triangle extends Shape
    }
    void printArea()
    {
        System.out.println("Area of Triangle:"+(0.5*dim1*dim2));
    }
}
class Circle extends Shape
{
    void input()
    {
        System.out.println(x: "Enter dimensions of Circle(radius):");
        dim1=s.nextDouble();
    }
    void printArea()
    {
        System.out.println("Area of Circle:"+(3.14*dim1*dim1));
    }
}
class AbstractArea
{
    Run | Debug
    public static void main(String args[])
    {
        Rectangle r=new Rectangle();
        Triangle t=new Triangle();
        Circle c=new Circle();
        r.input();
        t.input();
        c.input();
        r.printArea();
        t.printArea();
        c.printArea();
    }
}
```

0 ⚠ 1 Java: Ready Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Java Signed out Go Live

Output:

```
PROBLEMS 9   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
PS C:\Users\BMSCECSE\Desktop\1BF24CS254> cd "c:\Users\BMSCECSE\Desktop\1BF24CS254\" ; if ($?) { javac Area.java } ; if ($?) { java Area }
Enter dimensions of Rectangle(length and breadth):
4
5
Enter dimensions of Triangle(base and height):
3
6
Enter dimensions of Circle(radius):
7
Area of Rectangle:20.0
Area of Triangle:9.0
Area of Circle:153.86
PS C:\Users\BMSCECSE\Desktop\1BF24CS254>
```

Program 5:

Implement Bank Account operations

C: > Users > Riya > Downloads > Bank.java > ...

```java
import java.util.Scanner;

class Account {
    String customerName;
    int accno;
    String accType;
    double balance;

    Scanner s = new Scanner(System.in);

    void getAccountDetails() {
        System.out.print(s: "Enter customer name: ");
        customerName = s.nextLine();
        System.out.print(s: "Enter account number: ");
        accno = s.nextInt();
        System.out.print(s: "Enter account balance: ");
        balance = s.nextDouble();
    }

    void deposit() {
        System.out.print(s: "Enter amount to deposit: ");
        double amount = s.nextDouble();
        balance += amount;
        System.out.println(x: "Amount deposited successfully!");
        System.out.println("Updated balance: " + balance);
    }

    void withdraw() {
        System.out.print(s: "Enter amount to withdraw: ");
        double amount = s.nextDouble();
        if (amount > balance) {
            System.out.println(x: "Insufficient balance!");
        } else {
            balance -= amount;
            System.out.println(x: "Amount withdrawn successfully!");
            System.out.println("Updated balance: " + balance);
```

0 ⚠ 2 Java: Ready Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Java Signed out Go Live

C: > Users > Riya > Downloads > Bank.java > ...

```java
class Account {
    void withdraw() {
            System.out.println(x: "Amount withdrawn successfully!");
            System.out.println("Updated balance: " + balance);
        }
    }

    void displayBalance() {
        System.out.println(x: "\n--- Account Details ---");
        System.out.println("Account Holder: " + customerName);
        System.out.println("Account Number: " + accno);
        System.out.println("Account Type: " + accType);
        System.out.println("Current Balance: " + balance);
    }
}

class SavingsAccount extends Account {
    double interestRate = 0.05;

    void computeInterest() {
        System.out.print(s: "Enter time period in years: ");
        double time = s.nextDouble();
        double interest = balance * interestRate * time;
        balance += interest;
        System.out.println("Interest added: " + interest);
        System.out.println("Updated Balance: " + balance);
    }
}

class CurrentAccount extends Account {
    double minBalance = 1000;
    double penalty = 100;

    void check() {
        if (balance < minBalance) {
            System.out.println(x: "Minimum balance not maintained!");
```

0 ⚠ 2 Java: Ready Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Java Signed out Go Live

C: > Users > Riya > Downloads > J Bank.java > ...

```java
class CurrentAccount extends Account {
    void check() {
        if (balance < minBalance) {
            System.out.println(x: "Minimum balance not maintained!");
            balance -= penalty;
            System.out.println("Service charge imposed: " + penalty);
            System.out.println("Updated Balance: " + balance);
        } else {
            System.out.println(x: "Minimum balance maintained.");
        }
    }
}

public class Bank {
    Run | Debug
    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);

        System.out.print(s: "Enter account type (savings/current): ");
        String type = s.nextLine().toLowerCase();

        if (type.equals(anObject: "savings")) {
            SavingsAccount sa = new SavingsAccount();
            sa.accType = "Savings";
            sa.getAccountDetails();

            while (true) {
                System.out.println(x: "\n--- Savings Account Menu ---");
                System.out.println(x: "1. Deposit");
                System.out.println(x: "2. Withdraw");
                System.out.println(x: "3. Compute Interest");
                System.out.println(x: "4. Display Balance");
                System.out.println(x: "5. Exit");
                System.out.print(s: "Enter your choice: ");
                int choice = s.nextInt();

```

⊗ 0 ⚠ 2 Java: Ready Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Java Signed out Go Live

C: > Users > Riya > Downloads > J Bank.java > ...

```java
public class Bank {
    public static void main(String[] args) {
                switch (choice) {
                    case 1 -> sa.deposit();
                    case 2 -> sa.withdraw();
                    case 3 -> sa.computeInterest();
                    case 4 -> sa.displayBalance();
                    case 5 -> {
                        System.out.println(x: "Exiting...");
                        System.exit(status: 0);
                    }
                    default -> System.out.println(x: "Invalid choice!");
                }
            }
        } else if (type.equals(anObject: "current")) {
            CurrentAccount ca = new CurrentAccount();
            ca.accType = "Current";
            ca.getAccountDetails();

            while (true) {
                System.out.println(x: "\n--- Current Account Menu ---");
                System.out.println(x: "1. Deposit");
                System.out.println(x: "2. Withdraw");
                System.out.println(x: "3. Check Minimum Balance");
                System.out.println(x: "4. Display Balance");
                System.out.println(x: "5. Exit");
                System.out.print(s: "Enter your choice: ");
                int choice = s.nextInt();

                switch (choice) {
                    case 1 -> ca.deposit();
                    case 2 -> ca.withdraw();
                    case 3 -> ca.check();
                    case 4 -> ca.displayBalance();
                    case 5 -> {
                        System.out.println(x: "Exiting...");
```

⊗ 0 ⚠ 2 Java: Ready Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Java Signed out Go Live

C: > Users > Riya > Downloads > Bank.java > ...

```java
public class Bank {
    public static void main(String[] args) {
            }
        } else if (type.equals(anObject: "current")) {
            CurrentAccount ca = new CurrentAccount();
            ca.accType = "Current";
            ca.getAccountDetails();

            while (true) {
                System.out.println(x: "\n--- Current Account Menu ---");
                System.out.println(x: "1. Deposit");
                System.out.println(x: "2. Withdraw");
                System.out.println(x: "3. Check Minimum Balance");
                System.out.println(x: "4. Display Balance");
                System.out.println(x: "5. Exit");
                System.out.print(s: "Enter your choice: ");
                int choice = s.nextInt();

                switch (choice) {
                    case 1 -> ca.deposit();
                    case 2 -> ca.withdraw();
                    case 3 -> ca.check();
                    case 4 -> ca.displayBalance();
                    case 5 -> {
                        System.out.println(x: "Exiting...");
                        System.exit(status: 0);
                    }
                    default -> System.out.println(x: "Invalid choice!");
                }
            }
        } else {
            System.out.println(x: "Invalid account type entered!");
        }
    }
}
```

0 ⚠ 2 Java: Ready Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Java Signed out Go Live

Output:

```
PS C:\Users\Admin\Desktop\1BF24CS254>  & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Admin\AppData\Roaming\Code\User\workspaceStorage\099b02178a966e8eb1514e4018670ad1\redhat.java\jdt_ws\1BF24CS254_37831adc\bin' 'Bank'
Enter account type (savings/current): savings
Enter customer name: riya
Enter account number: 2
Enter account balance: 10000

--- Savings Account Menu ---
1. Deposit
2. Withdraw
3. Compute Interest
4. Display Balance
5. Exit
Enter your choice: 1
Enter amount to deposit: 500
Amount deposited successfully!
Updated balance: 10500.0

--- Savings Account Menu ---
1. Deposit
2. Withdraw
3. Compute Interest
4. Display Balance
5. Exit
Enter your choice: 3
Enter time period in years: 2
Interest added: 1050.0
Updated Balance: 11550.0

--- Savings Account Menu ---
1. Deposit
2. Withdraw
3. Compute Interest
4. Display Balance
5. Exit
Enter your choice: 2
Enter amount to withdraw: 700
Amount withdrawn successfully!
Updated balance: 10850.0

--- Savings Account Menu ---
1. Deposit
2. Withdraw
3. Compute Interest
4. Display Balance
5. Exit
Enter your choice: 4

--- Account Details ---
Account Holder: riya
Account Number: 2
Account Type: Savings
Current Balance: 10850.0
```

```
1. Deposit
2. Withdraw
3. Check Minimum Balance
4. Display Balance
5. Exit
Enter your choice: 3
Minimum balance not maintained!
Service charge imposed: 100.0
Updated Balance: 400.0

--- Current Account Menu ---
1. Deposit
2. Withdraw
3. Check Minimum Balance
4. Display Balance
5. Exit
Enter your choice: 1
Enter amount to deposit: 700
Amount deposited successfully!
Updated balance: 1100.0

--- Current Account Menu ---
1. Deposit
2. Withdraw
3. Check Minimum Balance
4. Display Balance
5. Exit
Enter your choice: 2
Enter amount to withdraw: 100
Amount withdrawn successfully!
Updated balance: 1000.0

--- Current Account Menu ---
1. Deposit
2. Withdraw
3. Check Minimum Balance
4. Display Balance
5. Exit
Enter your choice: 4

--- Account Details ---
Account Holder: saanvi
Account Number: 4
Account Type: Current
Current Balance: 1000.0

--- Current Account Menu ---
1. Deposit
2. Withdraw
3. Check Minimum Balance
4. Display Balance
5. Exit
Enter your choice: 5
Exiting...
PS C:\Users\Admin\Desktop\1BF24CS254>
                                                    Ln 78, Col 18   Spaces: 4   UTF-8   CRLF
```

```
--- Savings Account Menu ---
1. Deposit
2. Withdraw
3. Compute Interest
4. Display Balance
5. Exit
Enter your choice: 5
Exiting...
PS C:\Users\Admin\Desktop\1BF24CS254> ^C
PS C:\Users\Admin\Desktop\1BF24CS254>
PS C:\Users\Admin\Desktop\1BF24CS254>  c:; cd 'c:\Users\Admin\Desktop\1BF24CS254'; & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' '
C:\Users\Admin\AppData\Roaming\Code\User\workspaceStorage\099b02178a966e8eb1514e4018670ad1\redhat.java\jdt_ws\1BF24CS254_37831adc\bin' 'Bank'
Enter account type (savings/current): current
Enter customer name: saanvi
Enter account number: 4
Enter account balance: 500

--- Current Account Menu ---
1. Deposit
2. Withdraw
3. Check Minimum Balance
4. Display Balance
5. Exit
Enter your choice: 3
Minimum balance not maintained!
Service charge imposed: 100.0
Updated Balance: 400.0

--- Current Account Menu ---
1. Deposit
2. Withdraw
3. Check Minimum Balance
4. Display Balance
5. Exit
Enter your choice: 1
Enter amount to deposit: 700
Amount deposited successfully!
Updated balance: 1100.0

--- Current Account Menu ---
1. Deposit
2. Withdraw
3. Check Minimum Balance
4. Display Balance
5. Exit
Enter your choice: 2
Enter amount to withdraw: 100
Amount withdrawn successfully!
Updated balance: 1000.0

--- Current Account Menu ---
1. Deposit
2. Withdraw
3. Check Minimum Balance
4. Display Balance
```

Program 6:

Implement Packages

C: > Users > Riya > Downloads > Student.java > ...

```java
package CIE;
import java.util.Scanner;
public class Student
{
    protected String usn=new String();
    protected String name=new String();
    protected int sem;
    public void inputStudentDetails()
    {
        Scanner s=new Scanner(System.in);
        System.out.println(x: "Enter usn:");
        usn=s.nextLine();
        System.out.println(x: "Enter name:");
        name=s.nextLine();
        System.out.println(x: "Enter semester:");
        sem=s.nextInt();

    }
    public void displayStudentDetails()
    {
        System.out.println("USN:"+usn);
        System.out.println("Name:"+name);
        System.out.println("Semester:"+sem);
    }
}
```

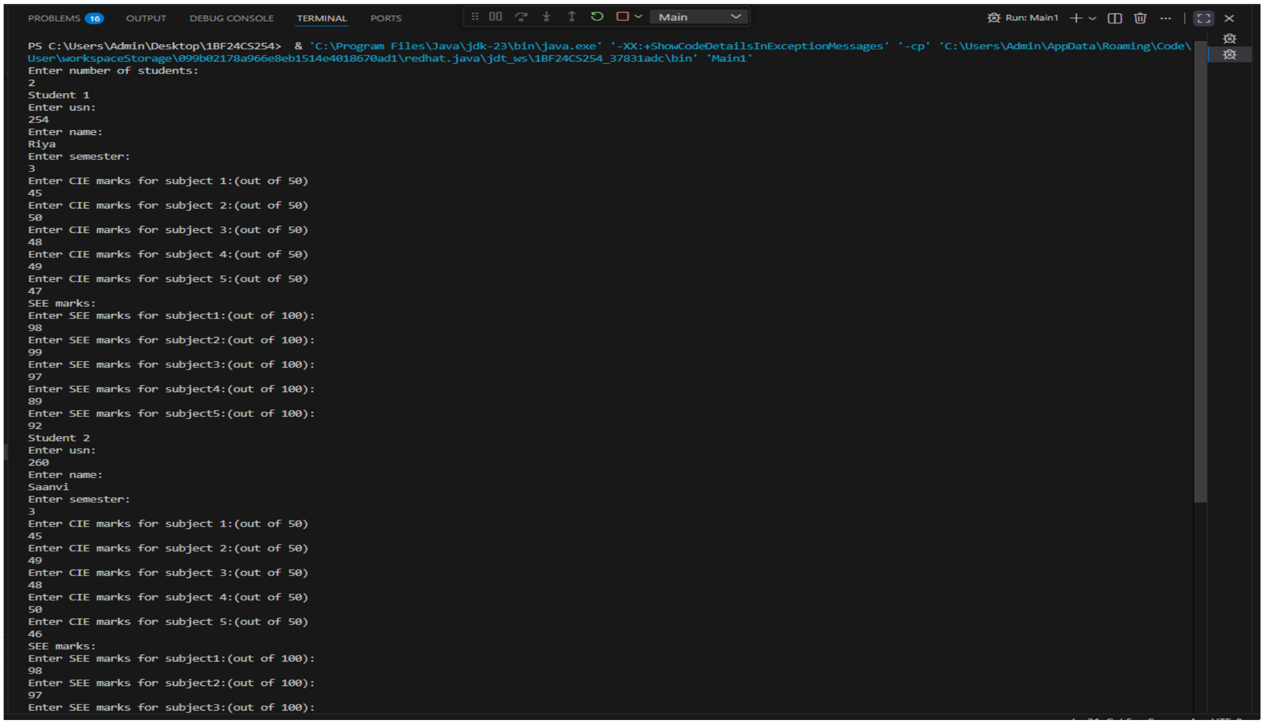
C: > Users > Riya > Downloads > Internals.java > ...

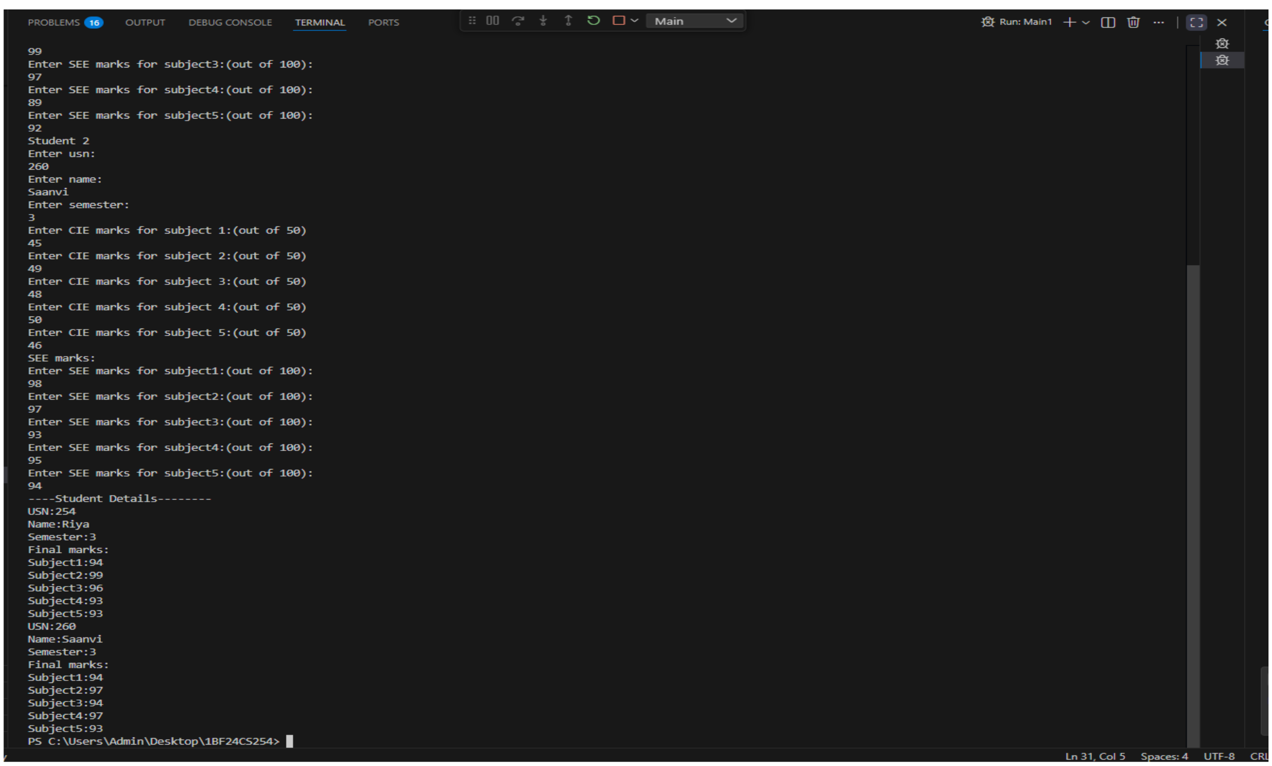
```java
package CIE;
import java.util.Scanner;
public class Internals extends Student
{
    protected int cmarks[]=new int[5];
    public void inputCIEmarks()
    {
        Scanner s=new Scanner(System.in);
        for(int i=0;i<5;i++)
        {
            System.out.println("Enter CIE marks for subject "+(i+1)+":(out of 50)");
            cmarks[i]=s.nextInt();
        }
    }
}
```

C: > Users > Riya > Downloads > Main1.java > ...

```java
import SEE.Externals;
import java.util.Scanner;

public class Main1 {
   public Main1() {
   }

   Run | Debug
   public static void main(String[] var0) {
      Scanner var1 = new Scanner(System.in);
      System.out.println(x: "Enter number of students:");
      int var2 = var1.nextInt();
      Externals[] var3 = new Externals[var2];

      int var4;
      for(var4 = 0; var4 < var2; ++var4) {
         System.out.println("Student " + (var4 + 1));
         var3[var4] = new Externals();
         var3[var4].inputStudentDetails();
         var3[var4].inputCIEmarks();
         var3[var4].inputSEEmarks();
         var3[var4].calculateFinalMarks();
      }

      System.out.println(x: "----Student Details--------");

      for(var4 = 0; var4 < var2; ++var4) {
         var3[var4].displayStudentDetails();
         var3[var4].displayFinalMarks();
      }

   }
}
```

Output:


PROBLEMS 16
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Main
Run: Main1
PS C:\Users\Admin\Desktop\1BF24CS254> & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Admin\AppData\Roaming\Code\User\workspaceStorage\099b02178a966e8eb1514e4018670ad1\redhat.java\jdt_ws\1BF24CS254_37831adc\bin' 'Main1'
Enter number of students:
2
Student 1
Enter usn:
254
Enter name:
Riya
Enter semester:
3
Enter CIE marks for subject 1:(out of 50)
45
Enter CIE marks for subject 2:(out of 50)
50
Enter CIE marks for subject 3:(out of 50)
48
Enter CIE marks for subject 4:(out of 50)
49
Enter CIE marks for subject 5:(out of 50)
47
SEE marks:
Enter SEE marks for subject1:(out of 100):
98
Enter SEE marks for subject2:(out of 100):
99
Enter SEE marks for subject3:(out of 100):
97
Enter SEE marks for subject4:(out of 100):
89
Enter SEE marks for subject5:(out of 100):
92
Student 2
Enter usn:
260
Enter name:
Saanvi
Enter semester:
3
Enter CIE marks for subject 1:(out of 50)
45
Enter CIE marks for subject 2:(out of 50)
49
Enter CIE marks for subject 3:(out of 50)
48
Enter CIE marks for subject 4:(out of 50)
50
Enter CIE marks for subject 5:(out of 50)
46
SEE marks:
Enter SEE marks for subject1:(out of 100):
98
Enter SEE marks for subject2:(out of 100):
97
Enter SEE marks for subject3:(out of 100):


PROBLEMS 16
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Main
Run: Main1
99
Enter SEE marks for subject3:(out of 100):
97
Enter SEE marks for subject4:(out of 100):
89
Enter SEE marks for subject5:(out of 100):
92
Student 2
Enter usn:
260
Enter name:
Saanvi
Enter semester:
3
Enter CIE marks for subject 1:(out of 50)
45
Enter CIE marks for subject 2:(out of 50)
49
Enter CIE marks for subject 3:(out of 50)
48
Enter CIE marks for subject 4:(out of 50)
50
Enter CIE marks for subject 5:(out of 50)
46
SEE marks:
Enter SEE marks for subject1:(out of 100):
98
Enter SEE marks for subject2:(out of 100):
97
Enter SEE marks for subject3:(out of 100):
93
Enter SEE marks for subject4:(out of 100):
95
Enter SEE marks for subject5:(out of 100):
94
----Student Details--------
USN:254
Name:Riya
Semester:3
Final marks:
Subject1:94
Subject2:99
Subject3:96
Subject4:93
Subject5:93
USN:260
Name:Saanvi
Semester:3
Final marks:
Subject1:94
Subject2:97
Subject3:94
Subject4:97
Subject5:93
PS C:\Users\Admin\Desktop\1BF24CS254>
Ln 31, Col 5
Spaces: 4
UTF-8

Program 7:

Implement Exception Handling

C: > Users > Riya > Downloads > ExceptionHandling.java > ...

```java
import java.util.Scanner;
class WrongAge extends Exception {
    WrongAge() {
        super(message: "Age Error!");
    }
    WrongAge(String msg) {
        super(msg);
    }
}
class InputScanner {
    Scanner s = new Scanner(System.in);
}
class Father extends InputScanner {

    int fatherAge;

    Father() throws WrongAge {

        System.out.print(s: "Enter Father's Age: ");
        fatherAge = s.nextInt();

        if (fatherAge < 0) {
            throw new WrongAge(msg: "Age cannot be negative");
        }
    }

    void displayFather() {
        System.out.println("Father's Age: " + fatherAge);
    }
}
class Son extends Father {

    int sonAge;

    Son() throws WrongAge {

```

0 ⚠ 1 Java: Ready Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} Java Signed out Go Live

C: > Users > Riya > Downloads > ExceptionHandling.java > ...

```java
class Son extends Father {

    int sonAge;

    Son() throws WrongAge {

        System.out.print(s: "Enter Son's Age: ");
        sonAge = s.nextInt();

        if (sonAge >=fatherAge) {
            throw new WrongAge(msg: "Son's age cannot be greater than or equal to Father's age");
        }
        else if (sonAge < 0) {
            throw new WrongAge(msg: "Son's age cannot be negative");
        }
    }

    void displaySon() {
        System.out.println("Son's Age: " + sonAge);
    }
}
class ExceptionHandling{

    Run | Debug
    public static void main(String[] args) {

        try {
            Son s = new Son();
            s.displayFather();
            s.displaySon();
        }
        catch (WrongAge e) {
            System.out.println("Exception Caught: " + e.getMessage());
        }
    }
}
```

0 ⚠ 1 Java: Ready Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} Java Signed out Go Live

Output:

```
PROBLEMS 18   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                         Run: ExceptionHandling

PS C:\Users\Admin\Desktop\1BF24CS254>  & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Admin\AppData\Roaming\Code\User\workspaceStorage\099b02178a966e8eb1514e4018670ad1\redhat.java\jdt_ws\1BF24CS254_37831adc\bin' 'ExceptionHandling'
Enter Father's Age: 50
Enter Son's Age: 65
Exception Caught: Son's age cannot be greater than or equal to Father's age
PS C:\Users\Admin\Desktop\1BF24CS254> ^C
PS C:\Users\Admin\Desktop\1BF24CS254>
PS C:\Users\Admin\Desktop\1BF24CS254>  c:; cd 'c:\Users\Admin\Desktop\1BF24CS254'; & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Admin\AppData\Roaming\Code\User\workspaceStorage\099b02178a966e8eb1514e4018670ad1\redhat.java\jdt_ws\1BF24CS254_37831adc\bin' 'ExceptionHandling'
Enter Father's Age: 45
Enter Son's Age: -6
Exception Caught: Son's age cannot be negative
PS C:\Users\Admin\Desktop\1BF24CS254> ^C
PS C:\Users\Admin\Desktop\1BF24CS254>
PS C:\Users\Admin\Desktop\1BF24CS254>  c:; cd 'c:\Users\Admin\Desktop\1BF24CS254'; & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Admin\AppData\Roaming\Code\User\workspaceStorage\099b02178a966e8eb1514e4018670ad1\redhat.java\jdt_ws\1BF24CS254_37831adc\bin' 'ExceptionHandling'
Enter Father's Age: -4
Exception Caught: Father's Age cannot be negative
PS C:\Users\Admin\Desktop\1BF24CS254> ^C
PS C:\Users\Admin\Desktop\1BF24CS254>
PS C:\Users\Admin\Desktop\1BF24CS254>  c:; cd 'c:\Users\Admin\Desktop\1BF24CS254'; & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Admin\AppData\Roaming\Code\User\workspaceStorage\099b02178a966e8eb1514e4018670ad1\redhat.java\jdt_ws\1BF24CS254_37831adc\bin' 'ExceptionHandling'
Enter Father's Age: 45
Enter Son's Age: 15
Father's Age: 45
Son's Age: 15
PS C:\Users\Admin\Desktop\1BF24CS254>

Ln 23, Col 42   Spaces: 4   UTF-8
```

Program 8:

Implement Thread

C: > Users > Riya > Downloads > Threadpgm.java > College

```java
class College extends Thread
{
    public void run()
    {
        try
        {
            for(int i=0;i<10;i++)
            {
                System.out.println(x: "BMS College Of Engineering");
                Thread.sleep(millis: 10000);
            }
        }
        catch(Exception e)
        {
            System.out.println(x: "Thread Interrupted");
        }
    }
}
class Dept extends Thread
{
    public void run()
    {
        try
        {
            for(int i=0;i<10;i++)
            {
                System.out.println(x: "CSE");
                Thread.sleep(millis: 2000);
            }
        }
        catch(Exception e)
        {
            System.out.println(x: "Thread Interrupted");
        }
    }
}
```

⊗ 0 ⚠ 1 Java: Ready Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Java Signed out Go Live

C: > Users > Riya > Downloads > Threadpgm.java > College

class College extends Thread
    public void run()

```java
            {
                System.out.println(x: "Thread Interrupted");
            }
        }
    }
class Dept extends Thread
{
    public void run()
    {
        try
        {
            for(int i=0;i<10;i++)
            {
                System.out.println(x: "CSE");
                Thread.sleep(millis: 2000);
            }
        }
        catch(Exception e)
        {
            System.out.println(x: "Thread Interrupted");
        }
    }
}
class Threadpgm
{
    Run | Debug
    public static void main(String args[])
    {
        College c=new College();
        c.start();
        Dept d=new Dept();
        d.start();
    }
}
```

Output:

```
PROBLEMS 23   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                Run: Threadpgm

PS C:\Users\Admin\Desktop\1BF24CS254>  & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Admin\AppData\Roaming\Code\User\w
orkspaceStorage\099b02178a966e8eb1514e4018670ad1\redhat.java\jdt_ws\1BF24CS254_37831adc\bin' 'Threadpgm'
BMS College Of Engineering
CSE
CSE
CSE
CSE
CSE
BMS College Of Engineering
CSE
CSE
CSE
CSE
CSE
BMS College Of Engineering
BMS College Of Engineering
BMS College Of Engineering
BMS College Of Engineering
BMS College Of Engineering
BMS College Of Engineering
BMS College Of Engineering
BMS College Of Engineering
PS C:\Users\Admin\Desktop\1BF24CS254>
```